Z-F-002130(Army-GL)
June 30, 1982

INTERIM FEDERAL SPECIFICATION

APPLESAUCE, TRAY PACK

This interim Federal specification was developed by the US Army Natick Research and Development Laboratories, Natick, MA 01760, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for change to the preparing activity at the address shown above.

1. SCOPE

1.1 Scope. This specification covers applesauce, in tray pack cans.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

TT-C-495	-	Coatings, Exterior, for Tinned Food Cans
PPP-B-636	-	Boxes, Shipping, Fiberboard

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, and other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC: Atlanta, GA; Chicago, IL; Kansas City, MO; Forth Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents, and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

FSC 8915

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Military Specification:

MIL-L-35078 - Loads, Unit: Preparation of Nonperishable Subsistence Items; General Specification for

Military Standards:

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars used in the Preparation of Canned Foods for the Armed Forces

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

U.S. Department of Health and Human Services

Federal Food, Drug and Cosmetic Act and Regulations Promulgated Thereunder (CFR parts 100-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. Department of Agriculture

U.S. Standards for Grades of Canned Applesauce

(Application for copies should be addressed to the Information Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

U. S. Standards of Condition of Food Containers

(Application for copies should be addressed to the Director, Food Quality Assurance Division, Agricultural Marketing Service, U.S. Department of Agriculture, South Building, 14th and Independence Avenue, SW, Washington, DC 20250.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Association of Official Analytical Chemists

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

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National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Suite 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL RESEARCH COUNCIL

Food Chemicals Codex

(Application for copies should be addressed to the Office of Director, National Academy of Sciences, National Research Council, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.4, 6.1 and 6.4).

3.2 General. The canned applesauce shall meet the requirements for natural color type, natural flavor type, sweetened type of pack, regular (or comminuted) style, U.S. Grade A of the United States Standards for Grades of Canned Applesauce except that the degrees Brix shall be from a minimum of 16.5° to a maximum of 23.0°.

3.3 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.3.1 Apples. Any variety of apples suitable for canning may be used. However, the apples must be the latest season's crop.

3.3.2 Sugar. Sugar shall be white, refined, granulated, cane or beet sugar, or a combination thereof. It shall meet the bacterial requirements of MIL-STD-900.

3.3.3 Salt. Salt shall be white, refined sodium chloride, non-iodized, with or without anticaking agent specific to the canning industry and used only in the level specified herein (see 3.4).

3.3.4 Acids Ascorbic, citric, or malic acids shall meet the requirements of the Food Chemicals Codex.

3.4 Processing. The addition of salt shall not exceed 0.08 percent and the level for degree Brix after addition of sugar shall be from a minimum of 16.5° to a maximum of 23.0°. The pH of the product shall be adjusted to 4.0 or less, if necessary, with food grade malic or citric acid. Prior to processing, to retard oxidation, 0.25 percent by weight of food grade ascorbic acid may be thoroughly blended with the product.

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3.5 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with product such as to conform to the following requirements:

a. The temperature of the applesauce at time of filling shall be not less than 190°F minimum.
b. The average net weight per can shall be 108 ounces.
c. The net weight in an individual can shall be 106 ounces minimum.
d. The filled can shall be hermetically sealed with a minimum of 10 inches of vacuum immediately after filling.
e. Immediately after sealing the can shall be inverted and cooled rapidly to a temperature that will inhibit further product change.

3.6 Plant qualification. The product shall be prepared, processed and packaged under sanitary conditions in accordance with good manufacturing practice. The product shall be processed in establishments meeting the sanitary requirements of the applicable government inspection agency.

3.7 Federal Food, Drug and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with the requirements of this specificaton. The contractor shall assure product compliance prior to submitting it to the USDA for any examination.

4.2 Inspection and acceptance service. Product acceptability shall be determined by the USDA. The USDA will determine the degree of acceptance service necessary to assure compliance with the requirements of this specification. The cost of grading and acceptance services performed by the USDA involving inspectionn, official documentation, and related services shall be borne by the contractor.

4.3 Plant qualification. The product shall be unacceptable if produced and stored in plants which do not meet the requirements specified in 3.6.

4.4 First article inspection. When a first article is required (see 6.1), it shall be examined for the defects specified in U.S. Standard for Grades of Canned Applesauce, and tested for the characteristic specified in 4.8.3. The presence of any defect or failure of any test shall be cause for rejection of the first article.

4.5 Component and material inspection. Components and materials shall be inspected in accordance with the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.6 Ingredient examination. Conformance of ingredients to identity and condition requirements shall be determined by examination of pertinent labels, markings, U.S. Grade certificates, invoices, certificates of analysis, or other valid documents. In addition each ingredient shall be examined organoleptically as necessary to determine conformance to the condition requirements. Any nonconformance to an identity or condition requirement shall be cause for rejection of the ingredient lot or any involved product.

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4.7 In-process examination. In-process examination shall be performed to determine conformance to the ingredient preparation, product processing, can interior coating, and vacuum sealing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation; certificates of conformance; or other valid documents shall be cause for rejection of the involved product.

4.8 Tray pack inspection. The inspection lot shall include only tray packs produced in one workshift. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

4.8.1 Net weight inspection. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing by 30) from the weight of each tray pack in the sample. If the average net weight is less than 108 ounces or if the net weight of any individual can is less than 106 ounces, the lot shall be rejected.

4.8.2 Product inspection. The finished product shall be inspected for conformance to the requirements in 3.2 in accordance with the provisions of the United States Standards for Grades of Canned Applesauce.

4.8.3 pH testing. Three tray packs shall be selected at random from the lot. The contents shall be blended in a Waring blender or equivalent to form the sample. The pH of the sample shall be determined by a pH meter calibrated to a pH of 4.0 at 20°C. Calibrate the pH meter using certified buffer solutions available commercially or prepare 0.05M acid potassium phthalate buffer solution as outlined in section 50.007 of the 13th Edition of the AOAC. Electrodes shall be rinsed with distilled water prior to testing the samples. If the pH of the sample is more than 4.0 at 20°C, the lot shall be rejected.

4.9 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the United States Standards for Condition of Food Containers.

4.10 Shipping container examination. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in construction, closure, and reinforcement shall be in accordance with the appendix of PPP-B-636. In addition, the following defects shall be classified as follows: Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible. Minor: (1) Other required markings missing, incorrect or illegible; (2) Interior packing with fiberboard liner or pads not as specified; (3) When applicable, reinforcement with nonmetallic strapping or tape is not used. Level C shipping containers shall be examined only for the marking defects heretofore specified.

4.11 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged in accordance with level A or C as specified (see 6.1).

5.1.1 Level A. A size 1001 by 1206 by 200 tray pack can shall be filled with 108 ounces of food product, sealed, and thermoprocessed as specifed in 3.5. The base plate of the can shall be electrolytic chromium-coated steel and shall be of sufficient weight and temper to protect the product during shipment and storage. The entire inside area of the can shall be coated. The can shall be coated overall on the outside with a coating conforming to type I of TT-C-495.

5.1.2 Level C. The product shall be packaged as specified in 5.1.1. except that cans with commercial exterior coating will be acceptable.

5.2 Packing. The product shall be packed in accordance with level A, B or C as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed, closed and reinforced in accordance with style RSC-L, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pads shall be full length and width less approximately 1/4 inch in each dimension. The box liner and pads shall be fabricated of grade V3c fiberboard.

5.2.2 Level B packing. Four cans of product, packaged as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

5.2.2.1 Nonmetallic reinforcement of shipping containers. When specified (see 6.1 and 6.3), the shipping container specified in 5.2.2 shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive, filament reinforced tape in accordance with the appendix of PPP-B-636.

5.2.3 Level C packing. Four cans of product, packaged as specified in 5.1, shall be packed in a manner to ensure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The shipping container shall be in accordance with the Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Unit loads. When specified (see 6.1), the product, packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified. For distribution to Navy, when unit loads are strapped, the strapping shall be limited to nonmetallic strapping.

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5.4 Labeling. Each unit pack shall be labeled to comply with the Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder. Paper labels are not permitted. The name of the product or abbreviation thereof and date of pack (day, month, year) shall be embossed or lithographed on the end (lid) of the can. The yield shall also be shown as follows:

YIELD

25 servings of approximately 4 ounces each.

5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When a first article is required (see 3.1, 4.4 and 6.4).
(c) Level of packaging and packing required (see 5.1 and 5.2).
(d) When nonmetallic reinforcement is required for level B packing (see 5.2.2.1).
(e) Type and class of unit load when unit loading is required (see 5.3).

6.2 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.3 Selection of nonmetallic reinforcement. The nonmetallic reinforcement specified in 5.2.2.1 should be selected for shipping containers that are destined for transfer as sea or other specific overseas operations.

6.4 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample consisting of three tray cans. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for inspection and approval of the first article.

Preparing activity:

Army - GL

Project No. 8915-A645

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

Z-F-002130(Army-GL) Applesauce, Tray Pack

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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